

Useful Resources

1. Data Structures and Algorithms (DSA)

- **LeetCode (Google Company Tag):** The gold standard for OA and interview prep. Focus on medium and hard problems tagged with Google from the last 6 months. Practice writing your solutions strictly in Java to build muscle memory for standard libraries like PriorityQueue and HashMap.
- **NeetCode (neetcode.io):** An incredible structured roadmap (NeetCode 150) that categorizes problems by pattern (e.g., Sliding Window, Two Pointers, Graphs). It provides excellent video explanations.
- **Cracking the Coding Interview (CTCI) by Gayle Laakmann McDowell:** A must-read textbook. The walkthroughs of time/space complexity and the core Java solution implementations are foundational for onsite rounds.
- **CSES Problem Set:** A great collection of algorithmic puzzles that heavily test dynamic programming and graph theory—common topics in Google interviews.

2. SQL and Database Management

- **LeetCode Database Problems:** Excellent for practicing complex queries. Filter for "Hard" and practice implementing Window Functions, CTEs, and Self-Joins.
- **PostgreSQL Official Tutorial (postgresqltutorial.com):** Since Google interviews often allow standard or PostgreSQL syntax, this is a fantastic resource for brushing up on specific advanced features, database normalization concepts, and exact syntax before the interview.
- **HackerRank SQL Domain:** Great for progressive learning, starting from basic aggregations and scaling up to complex relational queries.

3. CS Fundamentals (OS, CN, OOP)

- **GeeksforGeeks (CS Subjects):** The best quick-reference guide for OS (paging, scheduling), Computer Networks (OSI model, TCP/IP), and DBMS architectures.
- **Java OOP Cheat Sheets:** Brush up on interfaces, abstract classes, garbage collection, and multithreading (concurrency is a frequent topic in Google Java interviews).
- **Teach Yourself Computer Science (teachyourselfcs.com):** A heavier resource, but excellent if you need to deeply relearn operating systems or networking fundamentals from textbooks like *Operating Systems: Three Easy Pieces*.

4. System Design (For Mid-Level/L4 or high-achieving New Grads)

- **ByteByteGo (by Alex Xu):** Highly visual and digestible explanations of complex distributed systems, load balancing, and database sharding.

- **Grokking the System Design Interview (Educative.io):** A structured course that walks through designing massive systems like YouTube, Google Drive, and Twitter.
- **InfoQ and High Scalability:** Blogs that break down the real-world architectures of massive tech companies.

5. Mock Interviews and Behavioral Practice

- **Pramp (pramp.com):** A free platform for peer-to-peer mock interviews. Practicing speaking your thoughts out loud while writing code in a shared document is crucial for Google.
- **Interviewing.io:** Offers anonymous mock interviews with senior engineers from FAANG companies. It provides unfiltered, realistic feedback on your technical and communication skills.
- **The STAR Method Framework:** For the "Googlyness" rounds, keep a personal document mapping your past projects, hackathons, and academic leadership experiences to the Situation, Task, Action, Result format.